

Question Number	Answer	Additional guidance	Mark
8(a)	An answer that includes the following point: • correct reason for adaptation seen in the photograph (1)	mark response as a whole e.g. {thick / lots of} fur to {conserve heat / conserve energy / prevent heat energy loss / act as insulator / protection from cold large paws to walk on snow (coat) {colour / pattern / spots} for camouflage sharp teeth for {biting / tearing / eating / catching} {meat / animals / prey} {sharp / long} claws for {added traction / catching prey / climbing (mountains)} small ears to reduce heat loss large tail for {balance / to wrap around to reduce heat energy loss} ignore helps it to survive	(1)

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8(b)	An explanation that includes three of the following points: - analyse {the numbers of (different) species (in each habitat) / species richness} (1) - analyse {the number of individuals in each species / species evenness} (1) - calculate {index of diversity / diversity index} (1) - compare {D / index of diversity} values (for habitats in different areas) (1)	ACCEPT determine {genetic diversity / number of different alleles / heterozygosity index} ACCEPT the formula $D = \frac{N(N-1)}{\sum n(n-1)}$	(3)
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Question Number	Answer	Additional guidance	Mark
8(c)(i)	- correct subtraction (1) - correct calculated answer given to 1 dp (1)	<u>Example of calculation:</u> 7100-1200 = 5900 $(5900 \div 7100) \times 100 = (-)83.1 (\%)$ Allow ecf Correct answer with no working shown gains both marks	(2)

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8(c)(ii)	An answer that includes one of the following points: - reduction in habitat size (1) - decreased numbers of prey (1) {disease / hunting (by humans) / poaching} (1)	ACCEPT habitat {destruction/ loss} ACCEPT global warming if linked to unfavourable change in habitat ignore {global warming / climate change / change in the environment} unqualified ACCEPT increased competition {for food / territory} ignore killed unqualified	(1)
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Question Number	Answer	Additional guidance	Mark
8(d)(i)	- correct calculation (1) - correct answer in standard form (1)	<u>Example of calculation:</u> $(250 \div 5) \times 6.5 = 325$ line X = 6 to 7 mm (0.6-0.7cm) 0 to 250 km = 5 to 5.5 mm (0.5-0.55cm) accept 2.7×10^2 to 3.5×10^2 Correct answer with no working shown scores full marks	(2)

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* 8(d)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • consideration of fragmentation of snow leopard range / separation of different populations / geographical isolation / populations in {central / northern/western / different areas in Asia} • consideration of how the fragmentation developed • consideration of different {environment / conditions} in each of the different locations (northern/eastern/southern) • idea of (different) selection pressures / (new) selection pressure occurs • idea of genetic variation in population / some individuals have different alleles / mutations have occurred / mutations resulted in different alleles • idea that different {mutations / alleles / adaptations} will be advantageous in different habitats • natural selection occurs / survival of the fittest • individuals with advantageous {allele / mutation} will have an increased chance of {surviving / reproducing} • passing on {advantageous allele / mutation} to offspring / increasing frequency of (advantageous) allele in gene pool • example of an advantageous {allele / phenotype} linked to the {habitat / location} • consideration of differences {in gene pool / phenotypes / anatomical, behavioural and physical adaptations} developing in different snow leopard populations • idea that inbreeding could occur / consideration of founder effect • consideration of (little) reproductive isolation / different populations having different (mating) behaviour or breeding times • sub-species can still breed together to form fertile offspring (6)

Level 0	0	No awardable content
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.

Question Number	Answer	Additional guidance	Mark
8(d)(iii)	An answer that includes the following point: individuals (from different sub-species) can (still) breed together to produce fertile offspring (1)	ACCEPT they can {breed / interbreed / reproduce / mate} (together) to produce fertile offspring ignore they can produce fertile offspring	(1)